GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

SKUdesk

One agent that buys cheapest, helps you sell, and finds arbitrage. Only for the identical product.

If nothing is provably the same item, or no gap survives the costs, it says no. A person approves every spend.

Live: skudesk.lol

github.com/johnnyduo/SKUdesk_

THE PROBLEM

Price comparison breaks the moment two listings are not the same thing.

We searched Google Shopping for **Apple iPhone 16 Pro 256GB**. It returned 40 listings. Among them:

- Twenty labelled pre-owned, refurbished or restored
- Pro Max, Plus and 16e models with almost the same name
- Listings that never say the storage size
- Two AT&T listings at **$21.59** and **$25.23** against prices near $629

What goes wrong

A naive agent "finds" a 96% discount on a $21 iPhone and recommends buying it. The cheapest price is usually the wrong product, or not a real price.

Arbitrage and buying agents are only as good as their idea of "same".

HOW IT WORKS

One gate decides what counts as the same item. Everything else is arithmetic.

1 · SEARCH

Google Shopping via SerpApi, eBay API ready. Every paid call cached; monthly cap.

2 · IDENTITY GATE

Gemini compares each listing with the product. It must **quote** the brand and set number from both listings.

3 · PRICE SANITY

Same product, believable price? Far below median or below half of retail is set aside.

4 · ECONOMICS

Integer-cent landed cost, fees, shipping, margin floor. Verdict or an honest no.

5 · APPROVAL

The agent files a pending proposal. A person approves. No payment details held.

Buy, sell and arbitrage are three views of the same verified set of identical offers.

GEN AI, DONE CAREFULLY

The model is not trusted. Its evidence is checked.

Running the gate on real listings exposed a failure our hand-made eval missed: the model credited listings with a **"LEGO"** brand they never wrote, and passed generic blocks that reuse a set number.

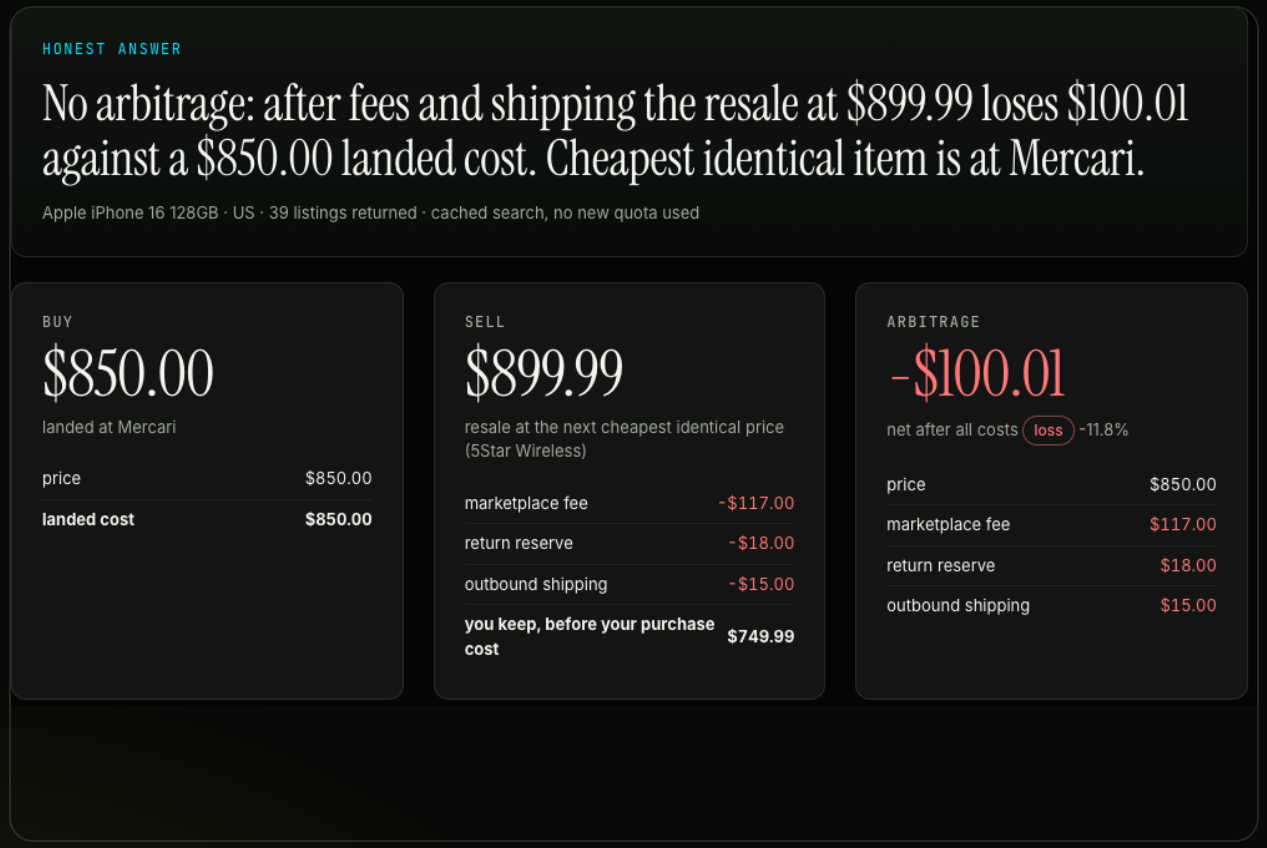
Fix: Gemini must copy the brand and identifier out of each listing. **Code verifies the quotes are really in the text**. Anything it cannot back up becomes "uncertain" and never passes. Errors fail closed.

Defences, all tested

- Verbatim evidence check on brand and identifier; storage and model tier read by code
- Photo check: Gemini looks at the picture and can only take a match away
- Listing text treated as untrusted (injection pairs in the eval)
- Price sanity against Gemini's recalled retail price, flagged as recalled, not live
- "Too good to be true" above a 50% margin

LIVE RESULT · IPHONE 16

The honest answer is often "no arbitrage". Here is why.



Ten listings reached the gate, **5 identical**, 5 different. Twenty-nine more were left out because Google labels them pre-owned or refurbished.

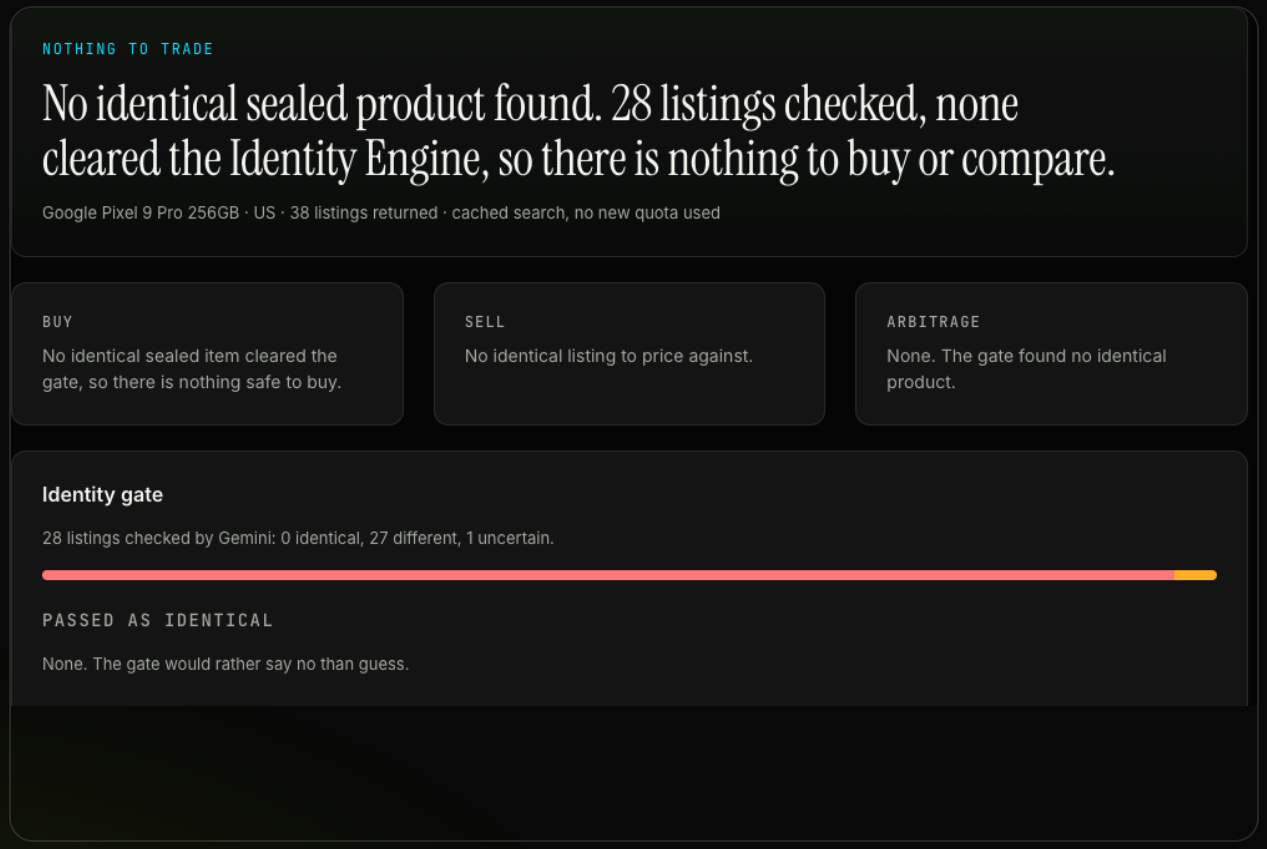
Gemini looked at each photo as well, and all five agreed with the text.

Cheapest real offer: **$850 landed**. Resale at the next price loses **$100.01**.

Asking prices, not sold prices. Fees, return reserve and shipping are shown and editable assumptions.

LIVE RESULT · PIXEL 9 PRO

When nothing is provably the same, the agent buys nothing.



28 listings for **Google Pixel 9 Pro 256GB**. **Zero** cleared the gate.

Pixel 9, 9 Pro XL, 9 Pro Fold and 128 GB units all carry nearly the same name. None is the 256 GB Pixel 9 Pro that was asked for.

No proposal is filed. Nothing is offered for approval.

EVIDENCE

60 hand-labelled pairs, run against real Gemini. The number that matters is zero.

0

false-identical: pairs that should not match but passed the gate.

56 of 60 exact. Three were labelled different and answered "uncertain", the safe side, and one identical pair was held back.

What the pairs cover

- Phones and LEGO sets: storage sizes, Pro against Pro Max, refurbished and carrier-locked units, cases, bundles, clones, injected instructions
- 5 titles are real ones captured live the day we found the leak
- Gemini also looks at each matching listing's photo and can only take a match away
- Barcode pairs are next

CONTROL AND SECURITY

The owner holds the controls. The agent can never spend.

Owner mode

- Per-buy limit and margin floor, enforced on every check
- A kill switch that stops scans, new proposals and approvals
- Watchlist for the daily Cloud Scheduler scan
- An audit log of every change and decision

Hardening

- Owner key in Secret Manager, compared in constant time
- Approvals need the requester's one-time token or the owner key
- Strict content policy, no inline script, untrusted listing text only ever set as text
- Rate limits, a monthly search budget, an allow-listed image proxy

BUILT ON GOOGLE CLOUD

Small, real and deployed. Plain web2 end to end.

PART	ROLE	STATE
Gemini	Identity gate, retail price recall	live
Cloud Run	API and one-page UI, deployed from source	live
Firestore	Cache, analyses, proposals, scan log	live
Cloud Scheduler	Daily watchlist scan, secret-protected	live
Secret Manager	API keys, least-privilege service account	live
eBay Browse API	Second source	written, not run against live API
Payments	Approval records the decision and opens the seller's link	by design

Smoke test: npm run smoke -- <url>

IMPACT AND NEXT

Retail teams lose money on "the same item" that was not.

- **Now:** new phones and sealed LEGO, US market, six markets selectable
- **Next:** GTIN evidence, eBay sold-price data, more categories where identity is checkable
- **Principle:** the agent's value is knowing when to say no, and showing its evidence when it says yes

Try it: skudesk.lol · **Code:** github.com/johnnyduo/SKUdesk_